

Luke Caffarey

Raleigh, NC | 919-695-6651 | lacaffar@ncsu.edu

linkedin.com/in/luke-caffarey-212473322 | github.com/lacaffar | lacaffar.github.io

Mechanical engineering student who designs, builds, and tests hardware end to end: CAD and FEA, parts I machine myself, integrated motors and sensors, and the test stands that prove the design. I train other students on every machine in the NC State senior design shop. Experience spans utility-scale wind, fusion diagnostics, and a student team I founded and lead.

EDUCATION

North Carolina State University, B.S. Mechanical Engineering, Economics minor

Expected May 2028

GPA 3.69 | Raleigh, NC | Coursework: thermodynamics, fluid mechanics, mechanics of materials, dynamics, machine design, Intro to Machine Learning, Aerodynamics (audit).

EXPERIENCE

GE Vernova, Engineering Intern, Fleet Performance Management

May to Aug 2026 | Greenville, SC

- Scoped and delivered five projects for the onshore wind fleet after arriving to an undefined role, sourcing the work directly from platform, controls, and reliability engineers.
- Built a Python tool that pulls drone-inspection imagery, reads blade serial numbers with a vision model, and reconciles them against fleet records; the inspection vendor had quoted \$50,000 for the same capability.
- Automated turbine controller fault triage in Python and SQL, routing qualifying flags to controls or platform engineering and saving 3 to 6 engineer-hours per week.
- Ran a measurement system analysis across two fleet databases to establish which source to measure forced controller variables from, and corrected 54 erroneous turbine configuration records.

NC State Mechanical and Aerospace Engineering, Senior Design Teaching Assistant

Aug 2026 to Present

- Train students on the full senior design machine shop: manual and CNC mills and lathes, drill press and saws, welding, waterjet and ProtoMAX, laser cutter, metal shear, Romer arm laser scanning, and Makerbot and Stratasys 3D printers.
- Coach capstone teams through design reviews, prototyping, and engineering drawing and documentation feedback.

FPAC Laboratory, NC State, Undergraduate Researcher

Dec 2024 to Apr 2025

- Analyzed spectroscopy and high-speed video from KSTAR fusion reactor experiments to support reactor wall stability assessment.
- Built Python time-series processing with adaptive Gaussian smoothing to improve peak detection in noisy diagnostic signals.

LEADERSHIP

Windpack, NC State Collegiate Wind Competition Team, Founder and President

Aug 2025 to Present

- Founded the team and grew it to 25+ members across mechanical, electrical, and project development sub-teams; led it to 6th of 10 at its first national competition in 2026.
- Leveraged my senior design lab TA position to train club members on the mills and lathes, moving the turbine off last year's 3D-printed parts and onto machined precision components this season.
- Broke from the field on blade prototyping: while every other team ran 3D-printed blades, we laid up a composite blade with inlaid glass fiber in epoxy, then ran a design trade study with a team of materials science engineering students to settle which construction is actually optimal.
- Helped lead the controls team in building a Simulink PID loop on an Arduino that feathers the blades and modulates generator resistance against wind speed and other measured inputs.
- Own the team's CAD and configuration practice in SolidWorks: assembly structure, part numbering, and interface definition between the mechanical and electrical sub-teams so their hardware integrates on first fit.

PROJECTS

Blade Dynamometer Test Stand, SolidWorks, ANSYS, and ESP32

2026

- Designed and built a benchtop dynamometer measuring torque, thrust, and RPM of blade prototypes, built because blades were the turbine's dominant failure mode last season and the team needed to test far more of them to failure before competition.
- Modeled vibration response in ANSYS and sized the frame and motor mounts with SolidWorks Simulation FEA to hold deflection and resonance out of the measurement path at peak rotor loading.
- Integrated load cell, RPM pickup, and motor controller on an ESP32 handling drive control, data acquisition, and logging.

ADDITIONAL EXPERIENCE

Wake County Government, EMS Division, EMS Cadet

Jan 2022 to May 2024 | Raleigh, NC

- Ran 12-hour shifts shadowing paramedics on 911 emergency response, assisting with patient care in high-pressure conditions.

SKILLS

CAD and Analysis: SolidWorks (part, assembly, drawings, Simulation FEA), ANSYS, QBlade, design for manufacturing

Fabrication: Manual and CNC mill and lathe, waterjet, laser cutter, metal shear and sheet metal, welding, FDM 3D printing, composite layup, soldering, Romer arm laser scanning for reverse engineering

Electronics: ESP32 and Arduino, Simulink PID control, load cells, encoders, IMU and barometer, motor controllers, data acquisition

Software: Python (signal processing, data analysis), MATLAB, SQL, C++ (embedded), Git, Linux and Raspberry Pi